



Multispectral Image Capture System

The Sixth Sensor: Jocelyn Ramirez, Chris Inderwiesche, Javier Hernandez, Jonnathan Terry, and Yu-Cheol Shin



The Problem

- Diagnosis procedures, home inspection, etc. are processes that can be invasive, time consuming, and costly but are necessary.
- Solution: The Sixth Sensor can discover hidden anomalies in a non-invasive way, by using infrared technology with a single click.

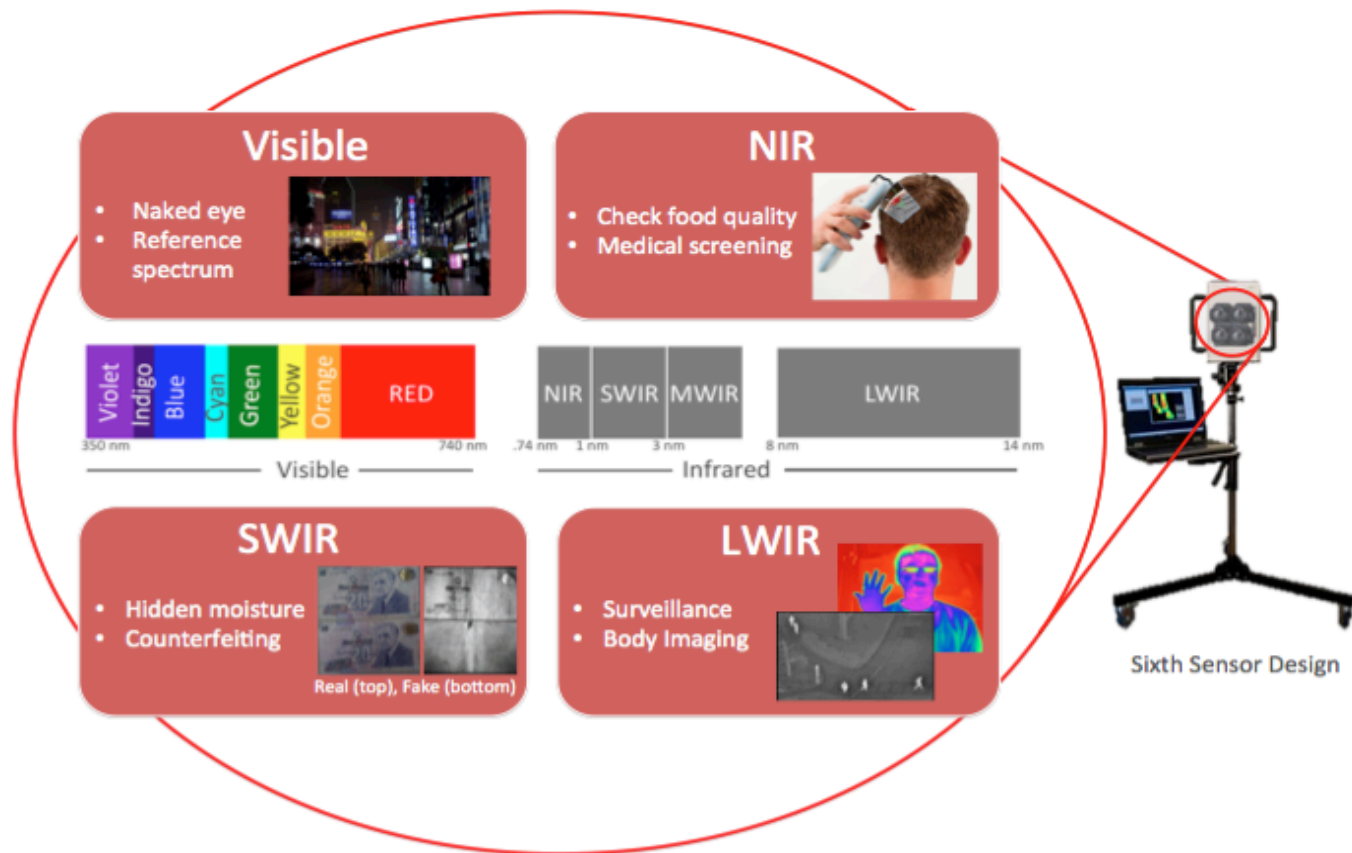


Overview

- Create a camera system that allows users to easily capture, annotate, and store images in four light wavelengths
- The camera system will allow users to discover the potential of infrared technology



Which Spectrums?



System Overview

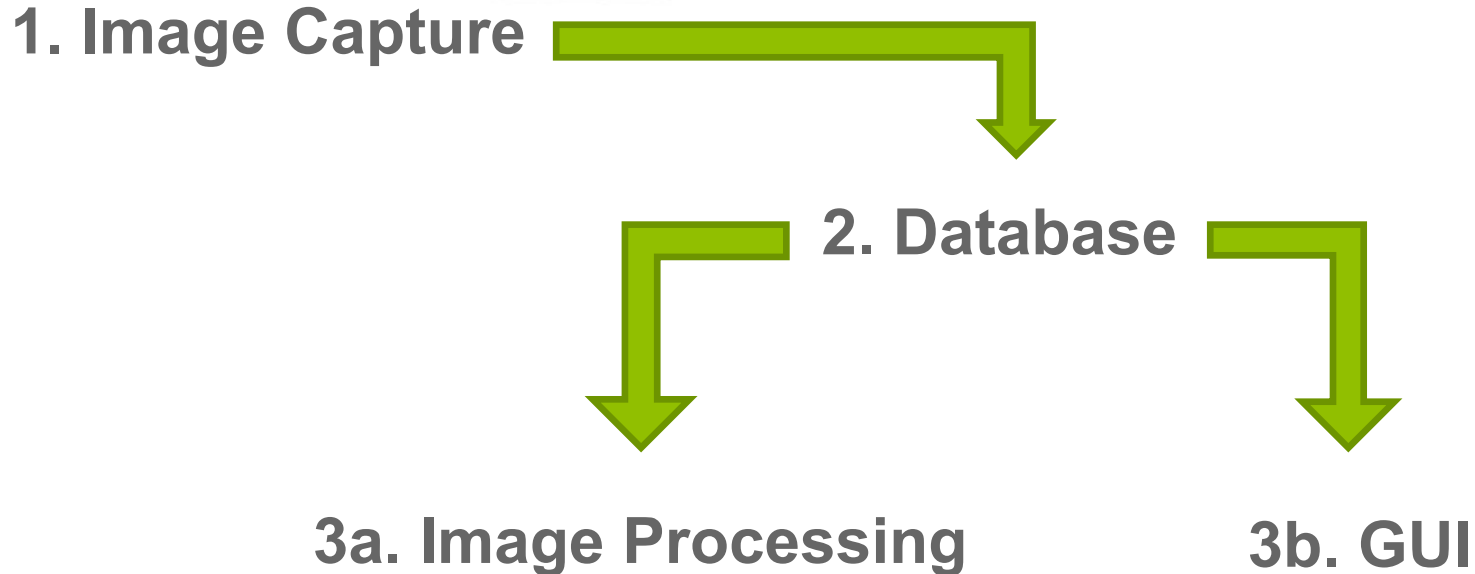
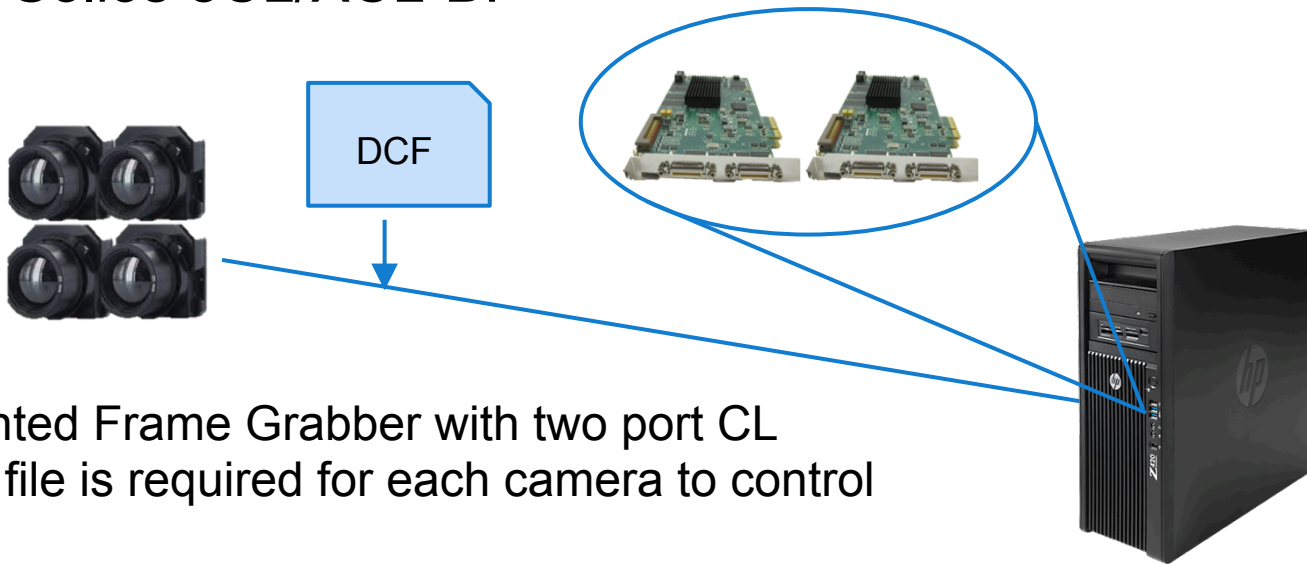


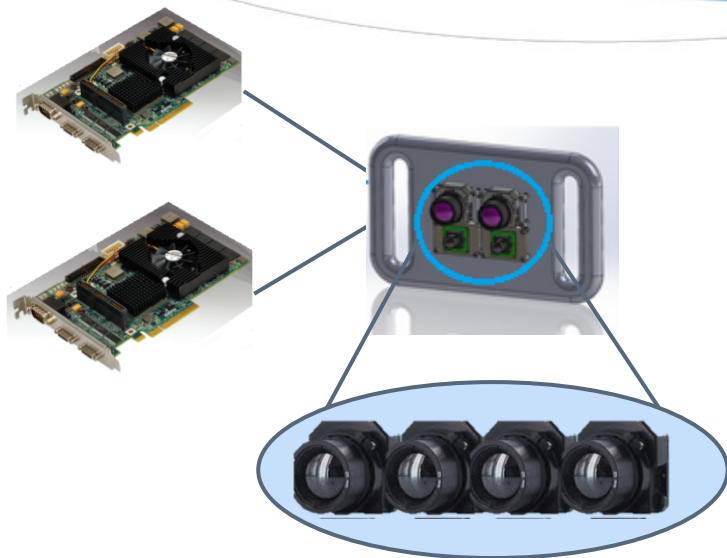
Image Capture

Matrox Solios eCL/XCL-B:



- Mounted Frame Grabber with two port CL
- DCF file is required for each camera to control

Image Capture



- Interface four different cameras with two frame grabbers
 - FLIR Tau SWIR
 - FLIR Tau LWIR
 - Imperx b1610 Monochrome (NIR)
 - Imperx b1610 Color (Visible)
- Features: Live Feed, Single Capture, and Save
- Synchronized capture of four images with a single button

Database

- Allow users to store images with a single click and retrieve existing images with no input necessary

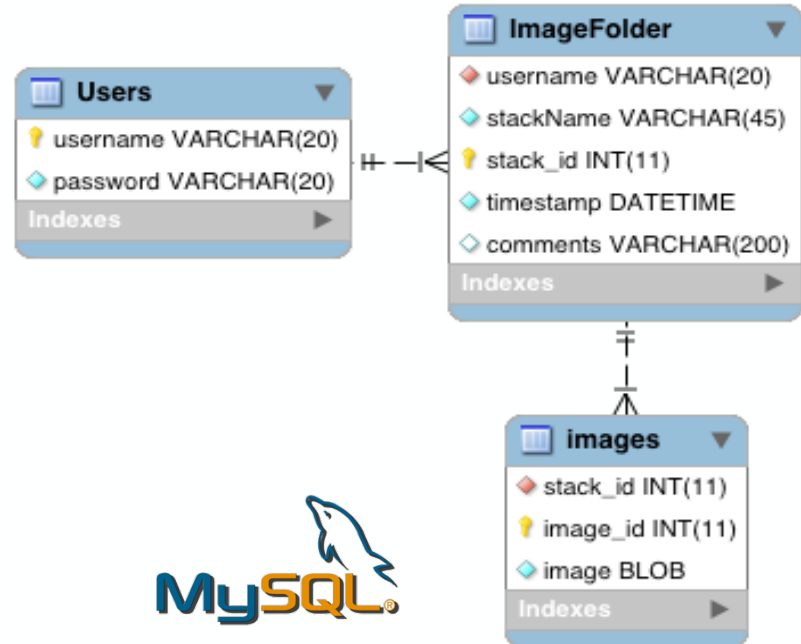


Image Processing: IrfanView

- IrfanView is a graphic viewer that contains features to manipulate images
- Reasons for using IrfanView
 - Create thumbnails
 - View image properties
 - Resample images to equal resolution size
 - Apply effects to the images

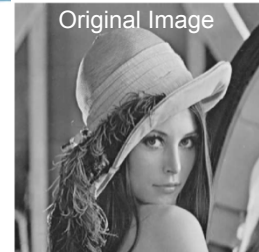


10101011
00110010
01110110
10000

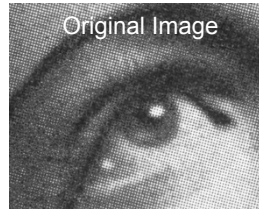


Image Processing: Tools

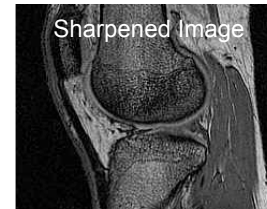
Edge Detection



Gaussian 'Smoothing'



Sharpening



GUI



FLIR



Username

Password

Login Screen



Image Stack Name	Upload Date	Comments
ex.	ex.	ex.

Image Folder Menu



**I
M
A
G
E

M
E
N
U**

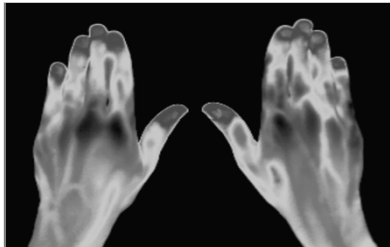


Image Analysis & Drawing Tools

Image Capture/Review

Demo

Questions?

Acknowledgments to UCSB, FLIR, and our advisors: Tim Sherwood,
Chandra Krintz, Kyle Jorgensen, and Janet Kayfetz